

IOT-ENABLED INTELLIGENT SEWER MONITORING AND FLOOD RISK EARLY-WARNING SYSTEM

A Project Report

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BONAFIDE CERTIFICATE

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ABSTRACT

Urban drainage and sewage systems frequently experience blockage, overflow, and inadequate water-flow management, particularly during heavy rainfall. These problems can result in waterlogging, flooding, environmental pollution, and health hazards. Conventional drainage monitoring methods mainly depend on manual inspection, which is time-consuming, labor-intensive, and unable to provide continuous real-time information. To address these limitations, this project proposes an Intelligent Drainage Blockage and Flood Prevention System based on Internet of Things (IoT) technology. The proposed system is designed to continuously monitor the condition of drainage and sewage water using multiple sensors integrated with an ESP8266-based control unit. An ultrasonic sensor is used to monitor the water level inside the drainage system, while a water-flow sensor measures the flow rate to identify abnormal or restricted flow conditions. A rain sensor is incorporated to detect rainfall conditions and support early identification of potential flood situations. The sensor data are processed by the microcontroller and transmitted to a web-based monitoring dashboard through wireless communication. The dashboard provides real-time information such as water level, water distance, flow rate, rainfall status, and overall system condition, enabling users to monitor the drainage system remotely. When the monitored parameters reach predefined critical levels or an abnormal flow condition is detected, the system can generate warning indications to alert the responsible personnel. This enables timely inspection and maintenance before severe blockage or flooding occurs. A laboratory-scale prototype has been developed to demonstrate the working principle of the proposed system, including water-level monitoring, flow measurement, and real-time dashboard visualization. The system provides a low-cost and scalable approach for smart drainage monitoring and can be further integrated with automated alert systems and larger municipal drainage networks. Overall, the proposed system aims to improve real-time drainage condition monitoring, early blockage detection, flood-risk identification, and preventive maintenance. By reducing dependence on manual inspection and providing continuous sensor-based information, the system can contribute to the development of safer, smarter, and more efficient urban drainage infrastructure.

Keywords: IoT, ESP8266, Smart Drainage, Sewage Water Monitoring, Water Level Monitoring, Flow Sensor, Rain Detection, Flood Prevention, Real-Time Monitoring.

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LIST OF SYMBOLS, ABBREVLATIONS AND NOMENCLATURE

1. SYMBOLS

SYMBOL	DESCRIPTION	UNIT
V	Supply Voltage	V
I	Current Consumed by the System	A
P	Electrical Power	W
D	Distance Measured by Ultrasonic Sensor	cm
H	Water Level in Drainage	cm
Q	Water Flow Rate	L/min
R	Rainfall Detection Value	—
T	Monitoring Time	s
Su	Ultrasonic Sensor Status	—
Sf	Water Flow Sensor Status	—
Sr	Rain Sensor Status	—
B	Blockage Detection Status	—
W	Warning Status	—
V _{cc}	Sensor Supply Voltage	V
V _o	Sensor Output Voltage	V
L	Drainage Water Level	%
t	Response Time	s

2. ABBREVIATIONS

ABBREVIATION	FULL FORM
IoT	Internet of Things
ESP8266	Espressif Systems 8266 Wi-Fi Microcontroller
LCD	Liquid Crystal Display
USB	Universal Serial Bus
DC	Direct Current
AC	Alternating Current
Wi-Fi	Wireless Fidelity
GPIO	General Purpose Input/Output
GND	Ground
VCC	Voltage At Common Collector / Supply Voltage
CSV	Comma-Separated Values
CAD	Computer-Aided Design
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
API	Application Programming Interface
IR	Infrared
LED	Light Emitting Diode
L/min	Litres Per Minute
cm	Centimetre
mm	Millimetre
%	Percentage

3. NOMENCLATURE

TERM	DEFINITION / DESCRIPTION
Intelligent Drainage System	System Designed to Monitor Drainage Conditions and Detect Possible Blockages and Flooding
Drainage	Channel or Pipeline Used to Carry Away Excess Water
Blockage	Accumulation of Waste or Material That Restricts Water Flow
Flood Prevention	Measures Used to Identify Drainage Problems Before Flooding Occurs
Water Level	Height Of Water Present Inside the Drainage System
Ultrasonic Sensor	Sensor Used to Measure the Distance Between the Sensor and the Water Surface
Water Flow Sensor	Sensor Used to Measure the Flow Rate of Water Through the Drainage System
Rain Sensor	Sensor Used to Detect the Presence and Intensity of Rainfall
ESP8266	Wi-Fi-Enabled Microcontroller Used to Collect Sensor Data and Communicate with the Dashboard
LCD Display	Display Used to Show Real-Time Sensor Readings and System Status
IoT Dashboard	Web-Based Interface Used to Monitor Sensor Data and System Warnings
Blockage Detection	Process Of Identifying Abnormal Reduction or Interruption in Drainage Water Flow
Water Flow Monitoring	Continuous Measurement and Observation of Water Flow Through the Drainage
Rain Detection	Process Of Identifying Rainfall Using a Rain Sensor Module
Warning Notification	Alert Generated When Abnormal Drainage Conditions Are Detected
Real-Time Monitoring	Continuous Observation of Drainage Parameters as They Change
Data Logging	Process Of Recording Sensor Measurements for Later Analysis
CSV Download	Feature Used to Download Recorded Monitoring Data in CSV Format

CHAPTER 1

INTRODUCTION

Urban drainage systems are essential for the safe and efficient management of wastewater and rainwater. However, the accumulation of solid waste, mud, plastic, and other debris can cause blockages and restrict water flow. During heavy rainfall, blocked drainage systems may lead to waterlogging, overflow, and localized flooding, creating environmental, infrastructure, and public health problems.

Traditional drainage inspection mainly depends on manual monitoring and periodic maintenance. This approach requires significant manpower and may not detect blockages or abnormal water levels at an early stage. Therefore, a continuous and real-time monitoring system is required to identify potential drainage problems before they result in severe flooding.

The proposed Intelligent Drainage Blockage & Flood Prevention System uses Internet of Things (IoT) technology for real-time drainage monitoring. An ESP8266 microcontroller is integrated with an ultrasonic sensor, water-flow sensor, and rain sensor. The ultrasonic sensor monitors the water level, the flow sensor measures water-flow conditions, and the rain sensor detects rainfall. The collected data are transmitted wirelessly to a web-based dashboard for real-time monitoring.

When abnormal conditions such as high-water level, reduced flow, or rainfall are detected, the system provides warning indications to support early action. This helps reduce dependence on manual inspection and enables timely maintenance. A low-cost prototype is developed to demonstrate the feasibility of IoT-based drainage monitoring and flood prevention.

Overall, the proposed system provides a simple, economical, and scalable solution for early drainage blockage detection, water-level monitoring, and flood-risk prevention. The system can be further developed for large-scale urban drainage networks with advanced alert and automated maintenance features.

1.1 Problem Statement

Drainage blockages caused by solid waste, mud, plastic, and other debris can significantly reduce water flow and increase the risk of overflow and flooding, especially during heavy rainfall. In conventional drainage systems, blockage detection and condition monitoring are mainly carried out through manual inspection or periodic maintenance. These methods are time-consuming, labour-intensive, and may not provide real-time information about the drainage condition.

A major challenge is the early identification of abnormal water levels and restricted water flow before they develop into serious drainage failures. The absence of continuous monitoring makes it difficult for maintenance personnel to take timely preventive action. In addition, rainfall conditions can rapidly increase the drainage water level, making early warning important for reducing flood risk.

Therefore, there is a need for a low-cost, real-time, and IoT-based drainage monitoring system that can continuously measure water level, water-flow conditions, and rainfall. The system should provide the collected information through a remote dashboard and generate warning indications when critical conditions are detected. This can support timely maintenance, reduce dependence on manual inspection, and improve the effectiveness of drainage blockage and flood prevention.

1.2 Objectives

The main objective of this project is to develop a low-cost IoT-based system for real-time drainage blockage monitoring and flood prevention. The specific objectives are:

- To monitor the water level inside the drainage system using an ultrasonic sensor.
- To measure water-flow conditions and identify possible flow restrictions using a flow sensor.
- To detect rainfall conditions using a rain sensor for early flood-risk assessment.
- To transmit sensor data wirelessly to a real-time monitoring dashboard using an ESP8266 microcontroller.
- To provide warning indications when critical water levels or abnormal flow conditions are detected.
- To support early maintenance and preventive action by reducing dependence on manual drainage inspection.

1.3 Scope of the Project

The scope of this project is to develop a low-cost IoT-based system for real-time monitoring of drainage conditions and early flood-risk detection. The system focuses on monitoring key parameters such as water level, water flow, and rainfall using suitable sensors connected to an ESP8266 microcontroller.

The collected sensor data are transmitted wirelessly to a web-based dashboard, where the drainage condition can be monitored in real time. The system provides warning indications when abnormal water levels, reduced flow, or rainfall conditions are detected. The prototype is intended to

demonstrate the feasibility of continuous drainage monitoring and support timely preventive maintenance.

The project is mainly focused on monitoring, early warning, and preventive action rather than automatic physical removal of drainage blockages. The developed concept can be further expanded for large-scale drainage networks by integrating advanced communication, automated alerts, and blockage-removal mechanisms.

1.4 Need for the Proposed System

The increasing occurrence of drainage blockages and waterlogging creates a need for an effective and continuous monitoring system. Conventional inspection methods depend mainly on manual observation and periodic maintenance, which may result in delayed detection of blockages and abnormal water levels. This can increase the risk of drainage overflow and localized flooding during heavy rainfall.

The proposed system is needed to provide real-time monitoring of drainage conditions using IoT-based sensors. Continuous measurement of water level, water flow, and rainfall can help identify abnormal conditions at an early stage. The use of a web-based dashboard enables maintenance personnel to monitor the system remotely and respond quickly when warning conditions occur.

The system also provides a low-cost and scalable approach that can reduce dependence on frequent manual inspection. Early detection and timely maintenance can improve drainage efficiency, reduce flood risk, and support safer urban infrastructure. The proposed system therefore addresses the need for an economical, reliable, and real-time solution for intelligent drainage monitoring and flood prevention.

1.5 Methodology

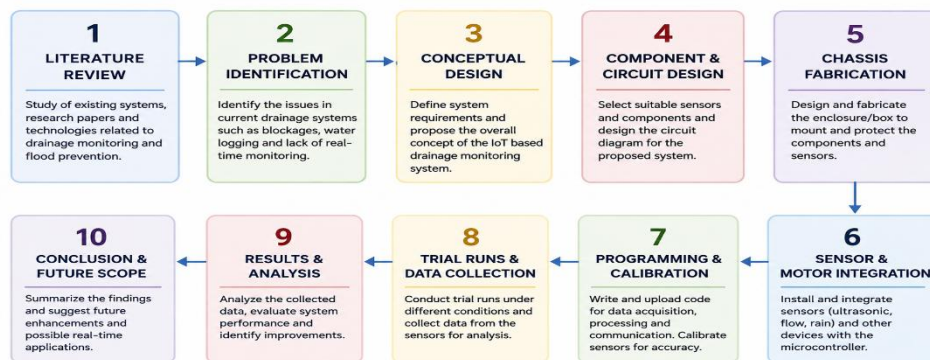


Figure 1.1 Methodology

The proposed system follows a systematic approach to monitor drainage conditions and provide

early warnings for blockage and flood risks. The methodology consists of sensor-based data collection, data processing, wireless communication, and real-time monitoring.

First, an ultrasonic sensor is installed to measure the water level inside the drainage system. A water-flow sensor is used to monitor the flow condition, while a rain sensor detects rainfall. These sensors continuously collect data from the drainage environment and send the readings to the ESP8266 microcontroller.

The ESP8266 processes the sensor data and compares the measured values with predefined threshold conditions. When an abnormal water level, reduced flow, or rainfall condition is detected, the system identifies the corresponding warning condition. The processed data are then transmitted through Wi-Fi to a web-based monitoring dashboard.

The dashboard displays the sensor readings and overall drainage status in real time. Warning indications are provided when critical conditions are detected, allowing users to take timely preventive action. The prototype is tested under different water-flow and rainfall conditions to verify the monitoring and warning functions.

Thus, the methodology integrates sensing, processing, wireless communication, and real-time visualization to provide an economical solution for intelligent drainage monitoring and flood prevention.

CHAPTER 2

LITERATURE REVIEW

Drainage Monitoring System Using IoT (2017)

Sriram Venkata Sai Bharath, T. Sidharth, Sravani Kaviti, and Bhuvaneswari Balachander developed an IoT-based drainage monitoring system using an ultrasonic sensor. The system was designed to monitor water levels in underground drainage systems and identify possible blockage conditions. The study demonstrated the feasibility of using ultrasonic sensing and IoT technology for continuous drainage monitoring and reducing dependence on manual inspection.

Prototype of Drainage Water Level Monitoring System Using Internet of Things (IoT) Based Web (2022)

Jaya Sihombing, Arie Jaenul, Sinka Wilyanti, and Nadya Putri developed a web-based drainage water-level monitoring system using IoT technology. An ultrasonic sensor was used to measure changes in drainage water level, while a buzzer provided an alert when the water approached a critical level. The study demonstrated that combining water-level sensing with web monitoring can help maintenance personnel respond quickly during heavy rainfall.

Development of a Smart Sensing Unit for LoRaWAN-Based IoT Flood Monitoring and Warning System in Catchment Areas (2023)

The researchers developed a low-cost IoT flood monitoring and warning system using an HC-SR04 ultrasonic sensor and an Arduino microcontroller. The system measured flood levels and classified conditions into different risk levels. LoRaWAN communication was used to transmit real-time information to IoT platforms and provide early warnings. The study demonstrated the suitability of wireless sensor networks for remote flood monitoring and early-warning applications.

Real-Time Sanitary Sewer Blockage Detection System Using IoT (2024)

Nour Faris, Tarek Zayed, Ehsan Aghdam, Ali Fares, and Ahmad Alshami developed a real-time IoT-based system for detecting sanitary sewer blockages. Low-power level sensors and 4G telemetry were used to continuously monitor sewage levels. Decision rules and time-series analysis were applied to distinguish blockage events from normal operating conditions. The research demonstrated that real-time monitoring can help detect blockages before they cause sewer overflows.

Drainage Blockage Monitoring and Detection System Using IoT (2025)

P. V. Kashid, Piyush Bhagat, Tejal Kavthe, Snehal Dongare, and Sakshi Mohite proposed an IoT-based system for real-time drainage blockage monitoring. The system incorporated ultrasonic sensors for water-level measurement and vibration sensors for detecting possible obstructions. The collected data were processed through a cloud platform, with machine-learning techniques proposed for predicting potential blockages from historical data. The study highlighted the importance of proactive drainage maintenance.

IoT-Based Clog Detection System (2025)

Vanessa Estrera and Fe Yara developed an affordable IoT-based clog detection system for drainage applications. The system used an ultrasonic sensor to monitor the water level, with an increase in water level used as an indication of a possible clog. The study focused on providing a low-cost alternative to expensive commercial clog-detection systems and demonstrated the potential of water-level monitoring for early blockage identification.

Design and Analysis of a Smart Drainage Alert System Using IoT Sensors (2025)

Shubh Agarwal developed a low-cost smart drainage alert system using IoT sensors for monitoring water levels and detecting possible drainage blockages. An ultrasonic sensor was connected to a microcontroller and cloud platform to provide live monitoring. When the water level exceeded a predefined threshold, an alert was generated for the user. The study demonstrated the usefulness of threshold-based IoT monitoring for affordable flood-risk management.

Dual-Purpose and Low-Cost IoT-Based Sensor Network for Real-Time Rainfall and Urban Flood Monitoring (2026)

Ismail Gul, Mohamed Abdelkader, Kyle Mackelburg, and Marouane Temimi developed a low-cost IoT sensor network capable of monitoring rainfall and surface-water levels simultaneously. The system used rainfall and ultrasonic water-level sensors to collect real-time data in flood-prone urban areas. The study demonstrated that combining rainfall and water-level information can provide improved monitoring and support timely decision-making during extreme weather events.

CHAPTER 3

COMPONENTS SPECIFICATION AND DESCRIPTION

3.1 COMPONENTS

The components used in the fabrication of the scaled prototype are listed in Table 3.1. Each component is described in detail in the following sections.

S.NO	COMPONENT	QUANTITY
1	ESP8266 NodeMCU	1
2	Ultrasonic sensor (HC-SR04)	1
3	Water Flow Sensor	1
4	5V power supply / Adapter	2
5	Rain Sensor	1
6	16x2 LCD display	1
7	Tap	1
8	Transparent box	1
9	Pipe	1
10	Jumper wires and connectors	1 set

Table 3.1 Components Used in the Prototype

3.2 COMPONENTS DESCRIPTION

3.2.1 ESP8266 NodeMCU

The ESP8266 NodeMCU is the main control unit of the proposed intelligent drainage monitoring system. It receives signals from the ultrasonic sensor, water-flow sensor, and rain sensor, processes the collected data, and transmits the information to the web-based monitoring dashboard through Wi-Fi. Its built-in wireless communication capability makes it suitable for IoT-based real-time drainage monitoring applications.

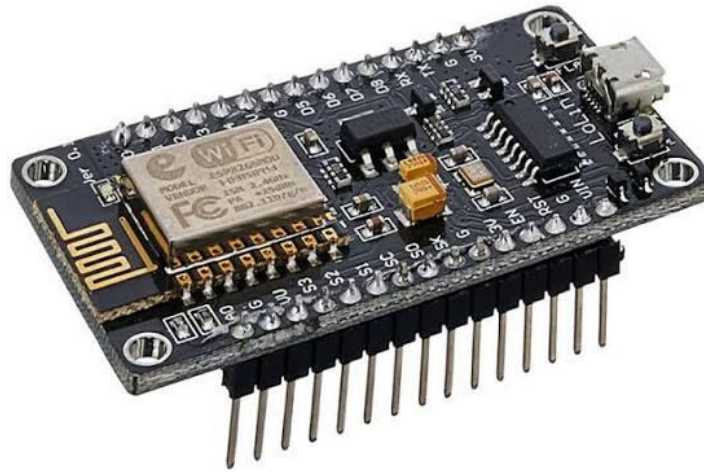


Figure 3.2.1 ESP8266 NodeMCU

PIN	FUNCTION
D1	Ultrasonic sensor TRIG
D2	Ultrasonic sensor ECHO
D5	Water-flow sensor signal
D6	Rain sensor digital output
A0	Rain sensor analog output
3V3	Sensor power supply
GND	Common ground

Table 3.2.1 Pin Configuration of Arduino UNO

3.2.2 Ultrasonic Sensor (HC-SR04)

The HC-SR04 ultrasonic sensor is used to measure the water level inside the drainage prototype. It operates by transmitting ultrasonic waves and measuring the time taken for the reflected wave to return. The measured distance is processed by the ESP8266 to determine the water level and identify conditions such as rising water or possible blockage.



Figure 3.2.2 Ultrasonic Sensor

PIN	FUNCTION
VCC	5V power supply
TRIG	Trigger signal from ESP8266
ECHO	Echo signal to ESP8266
GND	Common ground

Table 3.2.2 Pin Configuration of HC-SR04

3.2.3 Water Flow Sensor (YF-S201)

The **YF-S201 water-flow sensor** is used to measure the flow of water through the drainage prototype. It consists of an internal rotor and Hall-effect sensor that generates electrical pulses when water passes through the sensor. The pulse signal is connected to the ESP8266 microcontroller, which processes the pulses to estimate the water-flow rate. The measured flow data can be transmitted to the web-based dashboard for real-time monitoring.

The water-flow sensor is important for detecting **abnormal drainage conditions**. During normal operation, water flows continuously through the drainage channel, while a blockage may reduce the flow rate and cause the water level to increase. By monitoring both water flow and water level, the proposed system can identify possible drainage restrictions at an early stage and provide a warning for timely preventive action.



Figure 3.2.3 Water Flow Sensor

Pin/Wire	Function
Red	VCC power supply
Black	GND
Yellow	Pulse signal output

Table 3.2.3 Water Flow Sensor (YF-S201)

3.2.4 5V Power Supply / Adapter

The 5V power supply provides the required electrical power for the sensors and other electronic components of the prototype. A stable power source is important for maintaining reliable sensor operation and continuous monitoring. The power supply is connected to the appropriate power and ground terminals of the system.



Figure 3.2.4 5V Power Supply / Adapter

3.2.5 Rain Sensor Module

The rain sensor module is used to detect the presence and intensity of rainfall. The sensor provides an electrical output when water droplets fall on its sensing plate. The ESP8266 receives the sensor signal and uses it along with water-level and flow information to assess the possibility of drainage overflow and flooding.



Figure 3.2.5 Rain Sensor Module

3.2.6 LCD Display

The 16×2 LCD display is used to provide local visual information about the drainage monitoring system. It can display important parameters such as water level, water-flow condition, rainfall status, and overall system status. The display receives the processed information from the ESP8266 microcontroller and allows the user to observe the system condition directly without accessing the web dashboard.

The LCD display improves the usability of the proposed system by providing instant on-site information. It can also display warning messages when abnormal conditions such as high-water level, low water flow, or rainfall are detected. An I2C interface can be used to reduce the number of connections required between the LCD and ESP8266, making the circuit simpler and suitable for the prototype.



Figure 3.2.6 LCD Display

Pin/Wire	Function
Red	VCC Power Supply
Black	GND
Yellow	Pulse Signal Output

Table 3.2.6 LCD Display

CHAPTER 4

SYSTEM DESIGN AND WORKING PRINCIPLE

4.1 SYSTEM DESIGN

The proposed Intelligent Drainage Blockage & Flood Prevention System is designed to continuously monitor the condition of a drainage system using IoT technology. The system consists of an ESP8266 NodeMCU, ultrasonic sensor, water-flow sensor, rain sensor, LCD display, and software-based monitoring dashboard. The sensors collect information from the drainage environment and send the measured values to the ESP8266 for processing.

The ultrasonic sensor measures the water level inside the drainage channel, while the water-flow sensor monitors the flow of water. The rain sensor detects the presence of rainfall. The ESP8266 processes these sensor readings and determines the current drainage condition based on predefined threshold values. The processed information is displayed locally on the LCD and transmitted through Wi-Fi to the software dashboard for remote monitoring.

The software dashboard provides a clear representation of the drainage condition and displays parameters such as water level, water-flow rate, rainfall status, blockage condition, and flood warning. When the water level becomes high or the water flow decreases significantly, the system identifies a possible blockage or flood-risk condition and provides a warning to the user.

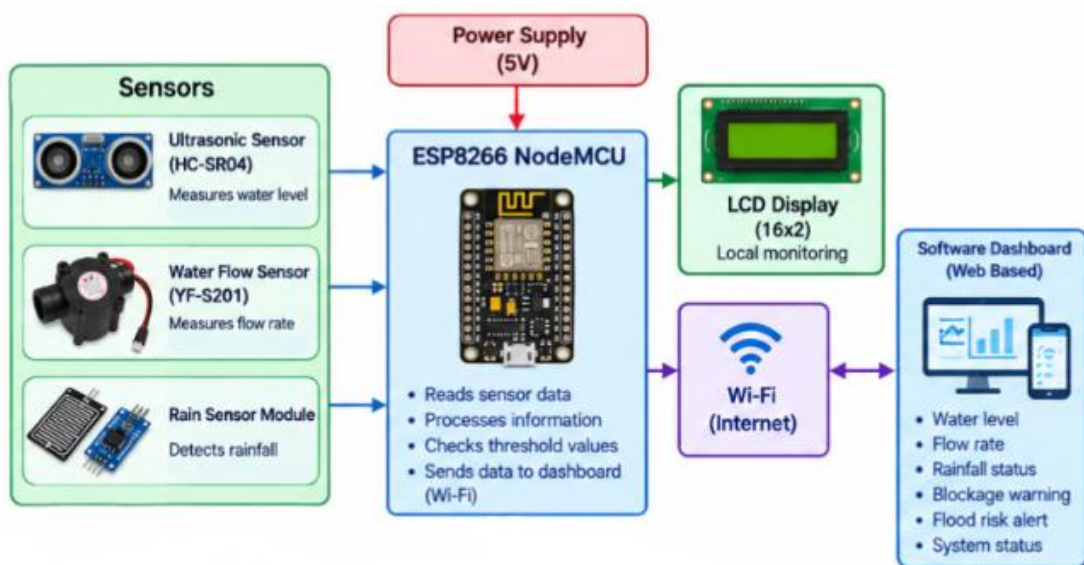


Figure 4.1 System Design

4.2 WORKING PRINCIPLE

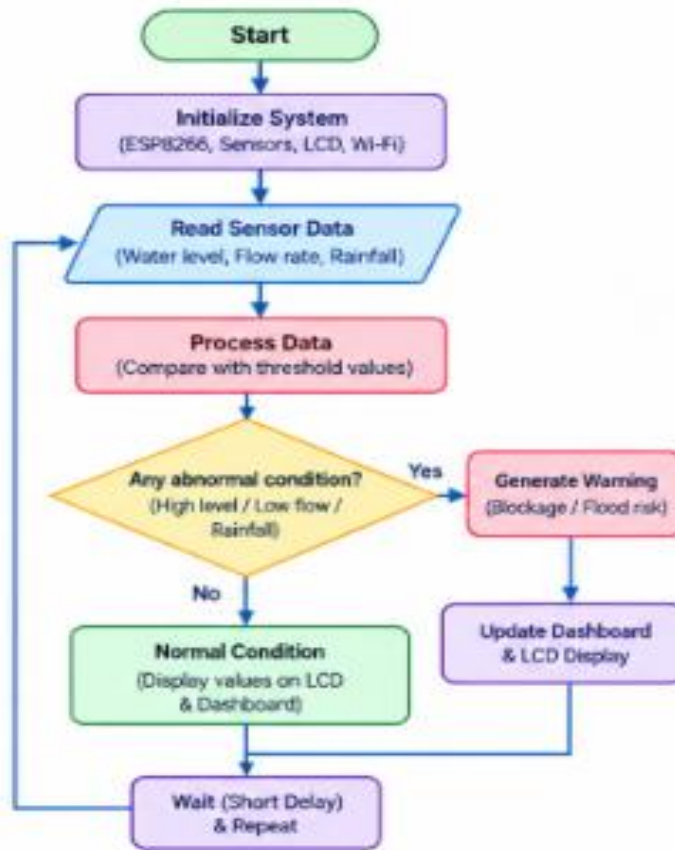


Figure 4.2 working principle

The system starts by continuously collecting data from the sensors installed in the drainage prototype. The ultrasonic sensor measures the distance between the sensor and the water surface, which is used to determine the water level. At the same time, the water-flow sensor generates pulses according to the movement of water, allowing the ESP8266 to estimate the flow rate. The rain sensor detects rainfall conditions and provides additional information for flood-risk assessment. The ESP8266 receives and processes all the sensor data. Under normal conditions, the system displays the measured values and indicates that the drainage condition is normal. If the water level increases beyond the predefined limit or the water-flow rate decreases, the system identifies a possible blockage. When rainfall is detected together with a rising water level, the system can indicate an increased flood risk. The processed data are displayed on the LCD display and transmitted through Wi-Fi to the software-based monitoring dashboard. The dashboard allows the user to monitor the drainage condition remotely and observe warning indications. Thus, the proposed system combines sensor monitoring, microcontroller-based processing, wireless communication, local display, and software-based visualization to provide early detection of drainage blockage and flood-risk conditions.

CHAPTER 5

COST ESTIMATION

This chapter presents an itemized cost estimation for the scaled prototype developed in this project, covering component procurement and fabrication. The figures represent indicative market prices at the time of this study.

5.1 COMPONENT COST

S.NO	COMPONENT	QUANTITY	PRICE (Rs.)
1	NodeMCU ESP8266	1	335
2	16×2 LCD Display with I2C	1	378
3	USB Micro-B Cable	1	282
4	Ultrasonic Sensor (HC-SR04)	1	374
5	Water Flow Sensor G1/2"	1	300
6	Water Level Sensor Module	1	74
7	Rain Sensor Module	1	100
8	5V Power Supply	1	250
9	Sensor Mounting/Prototype Box	-	200
	TOTAL ESTIMATED COST		2293

Table 5.1 Component Cost

5.2 FABRICATION COST

S.NO	ITEM	PRICE (Rs.)
1	Prototype Drainage Channel/Box	300
2	Sensor Mounting Arrangement	150
3	Protective Sensor Enclosure	200
4	Connecting & Supporting Materials	150
5	Fabrication and Assembly	300
	TOTAL FABRICATION COST	1100

Table 5.2 Fabrication cost

5.3 TOTAL PROJECT COST

S.NO	CATEGORY	PRICE (Rs.)
1	Component Sub-Total	2,170
2	Fabrication Sub-Total	650
3	Total Project Cost	2,820

Table 5.3 Total Project Cost

CHAPTER 6

INTELLIGENT DRAINAGE MONITORING WEBSITE DASHBOARD

6.1 WEBSITE OVERVIEW

The Intelligent Drainage Monitoring Website is designed to provide a simple and organized interface for monitoring drainage conditions in real time. The website displays important parameters such as water level, water flow rate, and rainfall status collected from the sensors connected to the ESP8266 NodeMCU. The system helps users observe drainage conditions remotely without the need for continuous manual inspection.

The dashboard provides a centralized platform for viewing sensor values, identifying abnormal drainage conditions, and receiving blockage and flood-risk warnings. This improves early detection of drainage problems and supports timely maintenance.

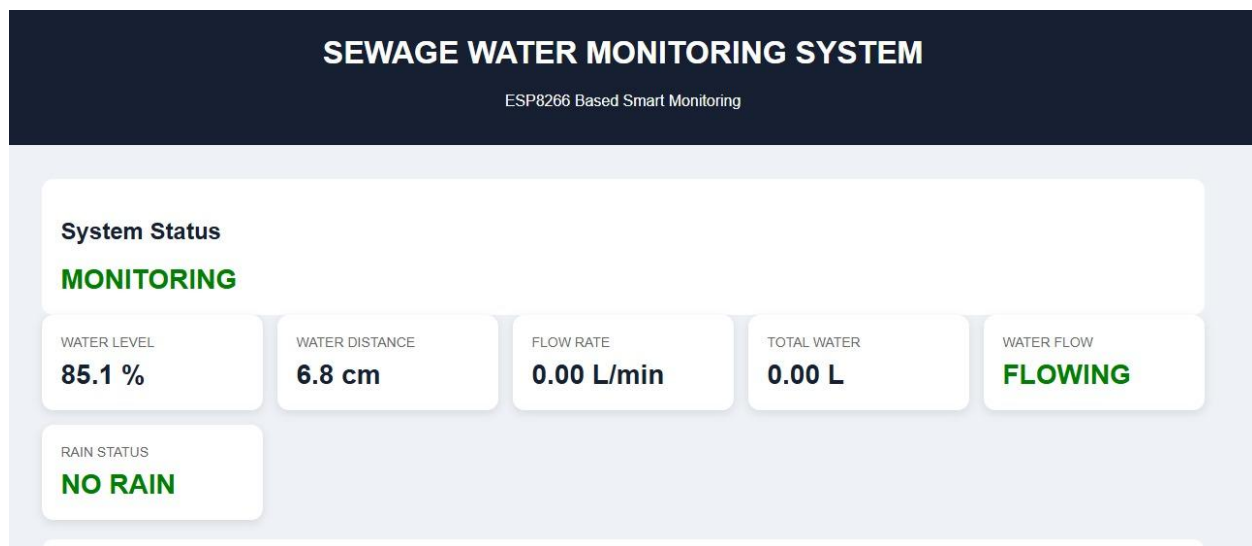


Figure 6.1 Website Overview

6.2 Rain Detection

The Rain Detection module displays the current rainfall condition detected by the rain sensor. The dashboard indicates whether rainfall is detected and helps the user understand the effect of rainfall on the drainage system.

Rainfall information can be observed along with water level and flow data to identify possible flood-risk conditions during heavy rainfall.

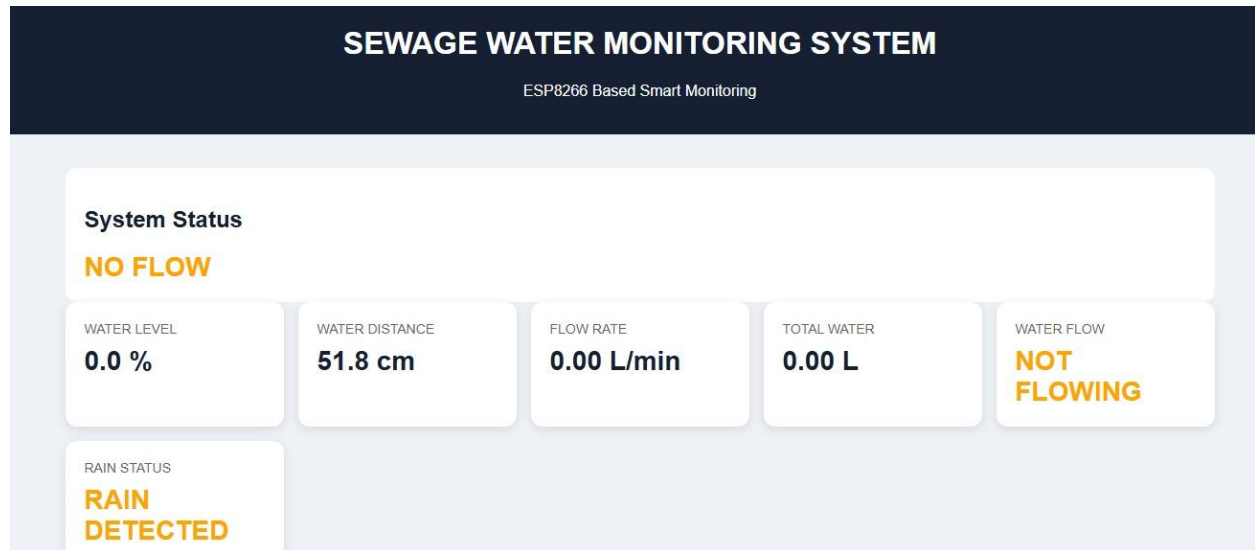


Figure 6.2 Rain Detection

6.3 Water Flow Monitoring

The Water Flow Monitoring module displays the flow condition of water inside the drainage channel. The water flow sensor provides flow information to the ESP8266, which transmits the processed data to the dashboard.

The dashboard can indicate different flow conditions such as Normal Flow and No Flow. A sudden reduction or absence of flow may indicate a possible blockage in the drainage channel.

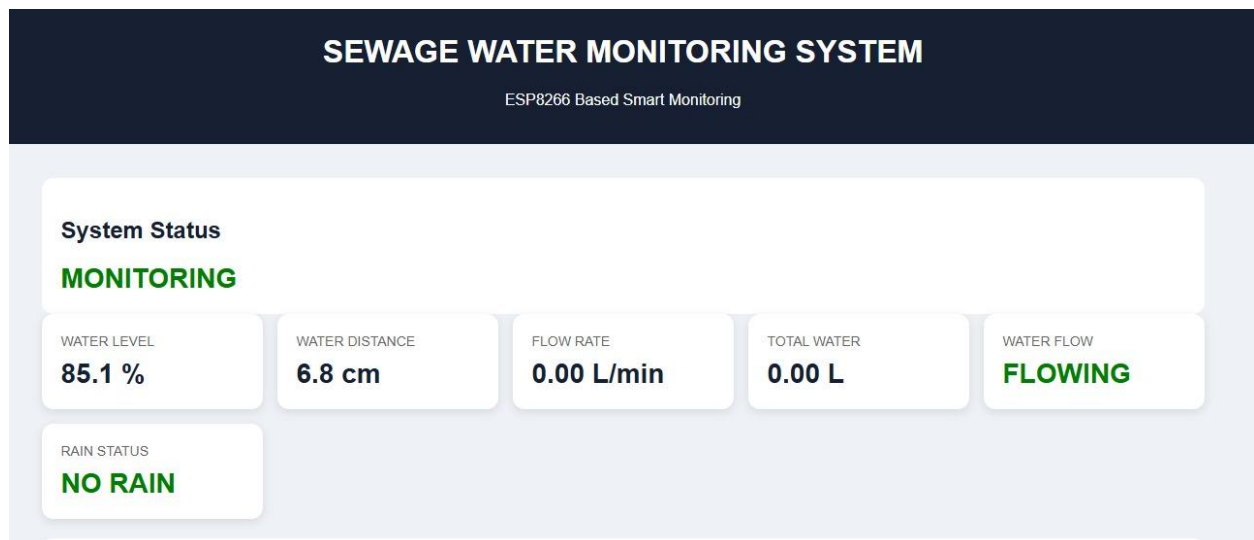


Figure 6.3 Water Flow Monitoring

6.4 Blockage Detection

The Blockage Detection module is used to identify possible restrictions in the drainage system. The system compares water level and flow conditions to determine whether the drainage is operating normally.

A high-water level combined with reduced or no water flow can indicate a possible blockage. When such a condition is detected, the dashboard displays a blockage warning to alert the user.

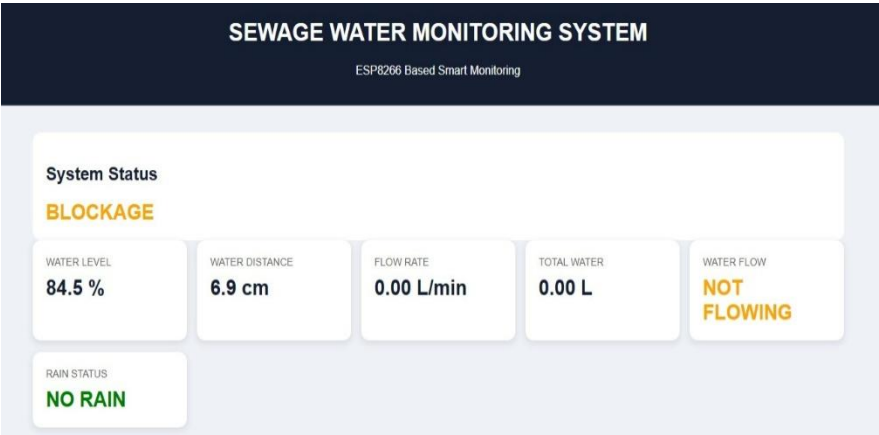


Figure 6.4 Blockage Detection

6.5 Live Water Monitoring Graph

The Live Water Monitoring section provides a graphical representation of the water level and flow conditions over time. The graph allows users to observe changes in drainage conditions continuously.

The live graph makes it easier to identify sudden increases in water level, reduction in flow, and possible overflow conditions. It also helps users analyse the performance of the drainage system during rainfall.



Figure 6.5 Live Water Monitoring Graph

6.6 Drainage Status Monitoring

The Drainage Status module provides a quick indication of the current condition of the drainage system. The system can display different operating conditions such as:

- Normal Flow – Water is flowing normally.
- No Flow – Water flow is absent or extremely low.
- Blockage Warning – Possible restriction is detected.
- Overflow Warning – Water level has reached a critical condition.
- Flood Risk – High water level and rainfall indicate a possible flood condition.

This status information allows users to understand the drainage condition quickly without analysing individual sensor readings.

6.7 Sensor Information

The Sensor Information section provides details about the sensors connected to the system. The dashboard can display the current readings and operating status of the ultrasonic sensor, water flow sensor, and rain sensor.

This section helps users verify whether the sensors are providing valid data and supports easy monitoring of the complete sensing system.

Sensor Information		
	Parameter	Value
	Water Level	0.0 %
	Distance	-1.0 cm
	Flow Rate	0.00 L/min
	Rain	NO
	Water Flow	FLOWING
	Status	NORMAL FLOW

Warning Information	
Overflow Level: 90%	
Blockage Detection Level: 65%	
Current Warning: NORMAL	

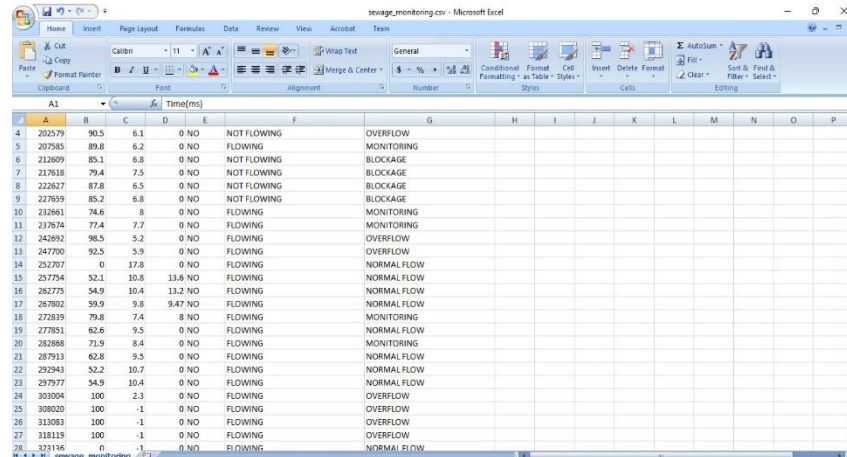
Download CSV

ESP8266 IP Address: 192.168.29.141

Figure 6.7 Warning Information

6.8 Data Monitoring and CSV Download

The dashboard provides a data management feature for recording and downloading sensor information. The collected values such as water level, water flow, rainfall status, and drainage condition can be stored with corresponding time information. The recorded data can be downloaded in CSV format for further analysis and documentation. The downloaded file can also be opened using spreadsheet software such as Microsoft Excel for generating graphs, reviewing historical data, and maintaining project records.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
4	202579	90.5	6.1	0	NO	NOT FLOWING	OVERFLOW									
5	207585	89.8	6.2	0	NO	FLOWING	MONITORING									
6	212609	85.1	6.8	0	NO	NOT FLOWING	BLOCKAGE									
7	217618	79.4	7.5	0	NO	NOT FLOWING	BLOCKAGE									
8	222627	87.8	6.5	0	NO	NOT FLOWING	BLOCKAGE									
9	227635	85.2	6.8	0	NO	NOT FLOWING	BLOCKAGE									
10	232661	74.6	8	0	NO	FLOWING	MONITORING									
11	237674	77.4	7.7	0	NO	FLOWING	MONITORING									
12	242692	98.5	5.2	0	NO	FLOWING	OVERFLOW									
13	247706	92.5	5.9	0	NO	FLOWING	OVERFLOW									
14	252707	0	17.8	0	NO	FLOWING	NORMAL FLOW									
15	257734	52.1	10.8	13.6	NO	FLOWING	NORMAL FLOW									
16	262775	54.9	10.4	13.2	NO	FLOWING	NORMAL FLOW									
17	267802	59.9	9.8	9.47	NO	FLOWING	NORMAL FLOW									
18	272839	79.8	7.4	8	NO	FLOWING	MONITORING									
19	277851	62.6	9.5	0	NO	FLOWING	NORMAL FLOW									
20	282868	71.9	8.4	0	NO	FLOWING	MONITORING									
21	287913	62.8	9.5	0	NO	FLOWING	NORMAL FLOW									
22	292943	52.2	10.7	0	NO	FLOWING	NORMAL FLOW									
23	297977	54.9	10.4	0	NO	FLOWING	NORMAL FLOW									
24	303004	100	2.3	0	NO	FLOWING	OVERFLOW									
25	308020	100	-1	0	NO	FLOWING	OVERFLOW									
26	313083	100	-1	0	NO	FLOWING	OVERFLOW									
27	318119	100	-1	0	NO	FLOWING	OVERFLOW									
28	323136	0	-1	0	NO	FLOWING	NORMAL FLOW									

Figure 6.8 Data Monitoring and CSV Download

CHAPTER 7

PROTOTYPE DEVELOPMENT

7.1 Concept and Design

The prototype was conceived as a compact drainage monitoring system capable of continuously monitoring water level, water flow, and rainfall conditions. The initial design phase involved planning the drainage channel layout, determining suitable sensor positions, and selecting components based on cost, availability, sensing accuracy, and compatibility with the ESP8266 NodeMCU platform.

The design was developed to provide real-time monitoring and early detection of abnormal drainage conditions such as no flow, blockage, and overflow. The sensor arrangement was planned to allow the ultrasonic sensor to measure water level, the water flow sensor to monitor flow conditions, and the rain sensor to detect rainfall.

7.2 Drainage Prototype Fabrication

The drainage prototype was fabricated as a compact model representing a small section of an urban drainage channel. Lightweight and easily available materials were used to construct the drainage channel and supporting structure. The prototype was designed with sufficient space for water flow and for mounting the required sensors.

The drainage channel includes suitable mounting positions for the ultrasonic sensor, water flow sensor, and rain sensor. The sensor locations were selected to provide clear measurement and avoid unnecessary interference with the water flow during testing.

7.3 Component Assembly

Following the fabrication of the drainage prototype, the ESP8266 NodeMCU, LCD display, sensors, and supporting electronic components were assembled and connected according to the system circuit design. The controller and display were positioned in a suitable location for easy access and observation during testing.

The sensor connections were routed carefully to maintain proper electrical connections and to keep the electronic components protected from water. The ESP8266 was connected to the monitoring dashboard through Wi-Fi for transmitting real-time sensor information.

7.4 Sensor Installation

The ultrasonic sensor was installed above the drainage channel and positioned towards the water surface to measure the water level. The water flow sensor was installed along the water path to detect the movement and flow condition of water through the drainage channel.

The rain sensor was positioned at the exposed section of the prototype to detect rainfall conditions. Proper sensor placement was maintained to obtain reliable readings during testing. The collected sensor data is processed by the ESP8266 and used to identify normal flow, no flow, blockage, and overflow conditions.

7.5 Final Prototype

The completed prototype consists of the drainage channel, ESP8266 NodeMCU, ultrasonic sensor, water flow sensor, rain sensor, and 16×2 I2C LCD display. The sensors are positioned at appropriate locations for monitoring the drainage condition, while the controller processes the collected information.

The final system is connected to the software-based monitoring dashboard, where live water-level and flow information, rainfall detection, sensor information, warning conditions, and monitoring graphs can be viewed. The recorded data can also be downloaded in CSV format for further analysis and documentation.

7.6 3D CAD MODELING

The 3D model was developed to represent the drainage monitoring system, including the drainage channel, sensor mounting positions, and electronic components. It was used to visualize the system layout before fabrication.

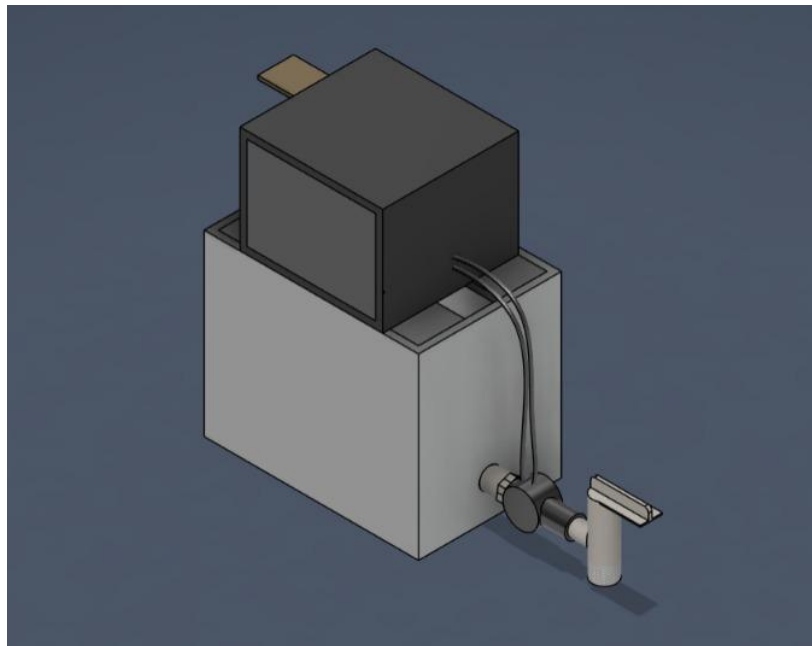


Figure 7.6 CAD Model

7.7 PROTOTYPE VIEWS

The final prototype was fabricated based on the 3D design. It integrates the drainage channel, sensors, ESP8266 controller, and LCD display for real-time monitoring and testing.

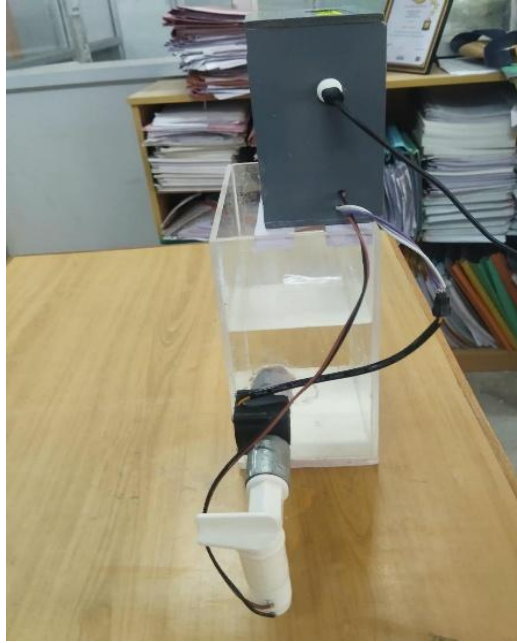


Figure 7.7.1 Prototype Side View



Figure 7.7.2 Prototype Front View

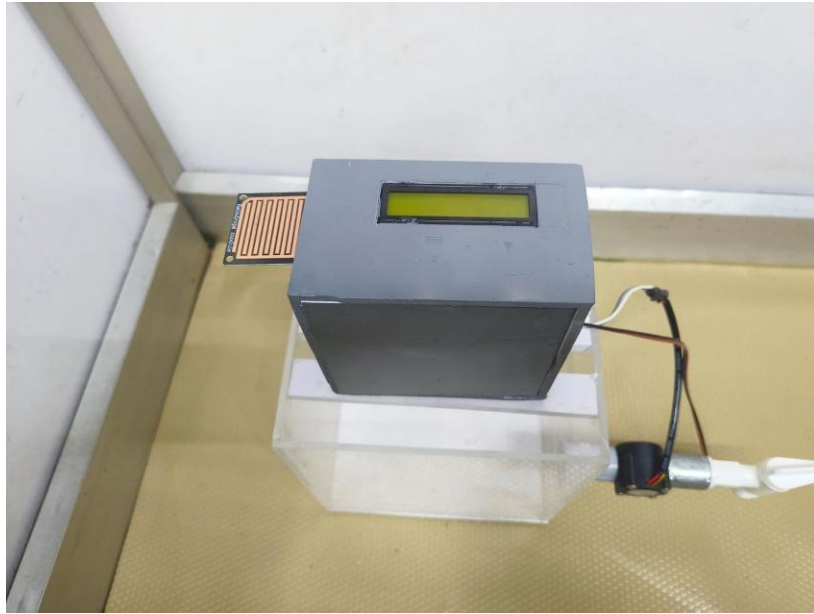


Figure 7.7.3 Prototype Top View

CHAPTER 8

RESULTS AND DISCUSSION

The completed drainage monitoring prototype was subjected to a series of controlled tests to evaluate its water-level measurement, water-flow detection, rainfall detection, blockage identification, overflow warning, and dashboard monitoring performance. Different drainage conditions were created during testing to verify whether the system could correctly identify normal flow, no-flow, blockage, and overflow conditions. The sensor readings were transmitted to the software dashboard for real-time observation and data recording.

8.1 Water Level Monitoring Test

The ultrasonic sensor was tested by varying the water level inside the drainage prototype. The measured water level was observed through the LCD and web dashboard. The system successfully detected changes in water level and generated an overflow warning when the water level reached the defined critical threshold.

Trial No.	Actual Water Level (cm)	Measured Water Level (cm)	Result
1	3	3.2	Pass
2	5	5.1	Pass
3	7	7.2	Pass
4	9	9.1	Pass
5	11	10.9	Pass

Table 8.1 Water Level Monitoring Test

8.2 Water Flow Detection Test

The water flow sensor was tested under different flow conditions to verify its ability to identify normal water movement and no-flow conditions. The sensor successfully detected water flow during normal operation and indicated a no-flow condition when the water movement was stopped.

Trial No.	Flow Condition	Sensor Status	Result
1	Normal Flow	Normal Flow	Pass
2	Normal Flow	Normal Flow	Pass
3	Reduced Flow	Low Flow	Pass
4	No Flow	No Flow	Pass
5	Normal Flow	Normal Flow	Pass

Table 8.2 Water Flow Detection Test

8.3 Rain Detection Test

The rain sensor was tested under different conditions to verify rainfall detection. Water droplets were applied to the sensor surface to simulate rainfall. The dashboard correctly indicated the change between dry and rainfall conditions.

Trial No.	Test Condition	Rain Detection	Result
1	Dry Condition	No Rain	Pass
2	Light Rain	Rain Detected	Pass
3	Moderate Rain	Rain Detected	Pass
4	Heavy Rain	Rain Detected	Pass
5	Dry Condition	No Rain	Pass

Table 8.3 Rain detection test results

8.4 Blockage Detection Test

A blockage was introduced into the drainage prototype to evaluate the ability of the system to identify abnormal drainage conditions. When the flow was restricted, the water level increased and the flow rate decreased. The system identified this combination as a possible blockage and displayed a blockage warning on the dashboard.

Trial No.	Drainage Condition	Water Level	Flow Status	Warning
1	Normal	Normal	Normal Flow	No Warning
2	Partial Blockage	Increased	Low Flow	Blockage Warning
3	Partial Blockage	Increased	Low Flow	Blockage Warning
4	Complete Blockage	High	No Flow	Blockage Warning
5	Normal	Normal	Normal Flow	No Warning

Table 8.4 Blockage detection test results

8.5 Overflow Detection Test

The overflow detection test was conducted by gradually increasing the water level in the drainage channel. When the water level reached the predefined critical limit, the system generated an overflow warning. This test confirmed that the prototype could provide an early indication of a potential overflow condition.

Trial No.	Water Level Condition	System Status	Result
1	Low	Normal	Pass
2	Medium	Normal	Pass
3	High	Warning	Pass
4	Critical	Overflow Warning	Pass
5	Critical	Overflow Warning	Pass

Table 8.5 Overflow detection test results

8.6 Dashboard and Live Monitoring Test

The software dashboard was tested to verify real-time display of sensor information. Water level, flow condition, rainfall status, blockage warning, and overflow warning were monitored during the test. The dashboard successfully displayed the changing sensor conditions and provided visual warning information.

The live water monitoring graph was also observed to verify changes in water level and flow conditions over time. Sensor information was recorded for further analysis and could be downloaded in CSV format for use in spreadsheet software such as Microsoft Excel.

Parameter	Monitoring Status	Result
Water Level	Live	Pass
Water Flow	Live	Pass
Rain Detection	Live	Pass
Blockage Status	Live	Pass
Overflow Status	Live	Pass
Warning Information	Displayed	Pass
CSV Data Download	Available	Pass

Table 8.6 Dashboard and Live Monitoring Test

8.7 Overall Performance

Taken together, the test results demonstrate that the proposed prototype can continuously monitor important drainage conditions using water-level, water-flow, and rainfall sensors. The system successfully identifies normal flow, no-flow, possible blockage, and overflow conditions and provides corresponding information through the LCD and web-based dashboard.

The integration of real-time monitoring, live graphs, warning information, and CSV data recording improves the usefulness of the proposed system for early drainage-condition detection. The prototype demonstrates the feasibility of a low-cost IoT-based approach for monitoring drainage systems and supporting timely maintenance before severe blockage or flooding occurs.

CHAPTER 9

SAFETY, ADVANTAGE AND APPLICATION

9.1 SAFETY FEATURES

The proposed drainage monitoring prototype incorporates basic safety features to ensure reliable and safe operation during testing. The ultrasonic sensor continuously monitors the water level, while the flow and rain sensors provide additional information about drainage conditions. The ESP8266 processes the sensor readings and generates warning indications when abnormal conditions such as high-water level, no flow, blockage, or overflow are detected.

The LCD display and web dashboard provide continuous status information to the user. Warning messages help identify critical drainage conditions at an early stage. The electronic components are protected from direct contact with water, and the sensor mounting arrangement is designed to provide stable and reliable measurements during prototype operation.

9.2 BATTERY AND ELECTRICAL SAFETY

The ESP8266, LCD, and sensor connections are supplied with suitable regulated power to prevent damage to the electronic components. All wiring connections should be properly insulated and secured to avoid short circuits, loose connections, or accidental contact with water. The power supply should be switched off before making any changes to the circuit.

Since the prototype operates near water, the electronic control unit should be kept inside a suitable protective enclosure. Water should not be allowed to enter the enclosure or come into direct contact with the ESP8266 and other electronic components. Sensors should also be inspected regularly for corrosion, damage, or improper mounting.

9.3 MAINTENANCE PROCEDURE

- Inspect all sensor connections and wiring periodically for loose connections or damage.
- Clean the ultrasonic sensor surface regularly to prevent dust, dirt, or water deposits from affecting water-level measurement.
- Check the water flow sensor for blockage or debris that may affect flow measurement.
- Clean the drainage prototype and remove unwanted particles that may interfere with sensor operation.
- Check the rain sensor periodically and ensure that its sensing surface is clean and properly exposed.
- Verify the ESP8266, LCD, and power supply connections before each testing session.
- Check and recalibrate sensor threshold values whenever sensor position or prototype conditions are changed.

- Monitor the dashboard regularly to verify that live sensor data and warning information are being updated correctly.
- Inspect the protective enclosure to ensure that water does not reach the electronic control components.

9.4 ADVANTAGES:

- Reduces manual effort involved in monitoring drainage conditions.
- Provides real-time monitoring of water level and flow conditions.
- Detects abnormal conditions such as blockage, no flow, and overflow.
- Provides rainfall detection for early flood-risk identification.
- Displays live monitoring information through the web dashboard.
- Provides warning information for critical drainage conditions.
- Stores monitoring data that can be downloaded in CSV format for analysis.
- Low-cost system suitable for prototype development and future IoT-based upgrades.

9.5 APPLICATIONS:

- Monitoring of urban drainage systems.
- Early detection of drainage blockage and overflow.
- Flood-risk monitoring in residential areas.
- Monitoring of stormwater drainage channels.
- Smart city and IoT-based drainage management systems.
- Monitoring of water flow in drainage networks.
- Rainfall and water-level monitoring applications.
- Research and demonstration of intelligent drainage monitoring systems.

CHAPTER 10

APPENDICES

APPENDIX – 1

10.1 CIRCUIT DIAGRAM

The circuit diagram shows the electrical connections of the intelligent drainage monitoring system. The ESP8266 NodeMCU acts as the main controller for all sensors and the LCD display. The ultrasonic sensor measures water level, while the flow sensor measures water flow. The rain sensor detects rainfall conditions in the drainage area. The LCD displays sensor values and warning information for real-time monitoring.

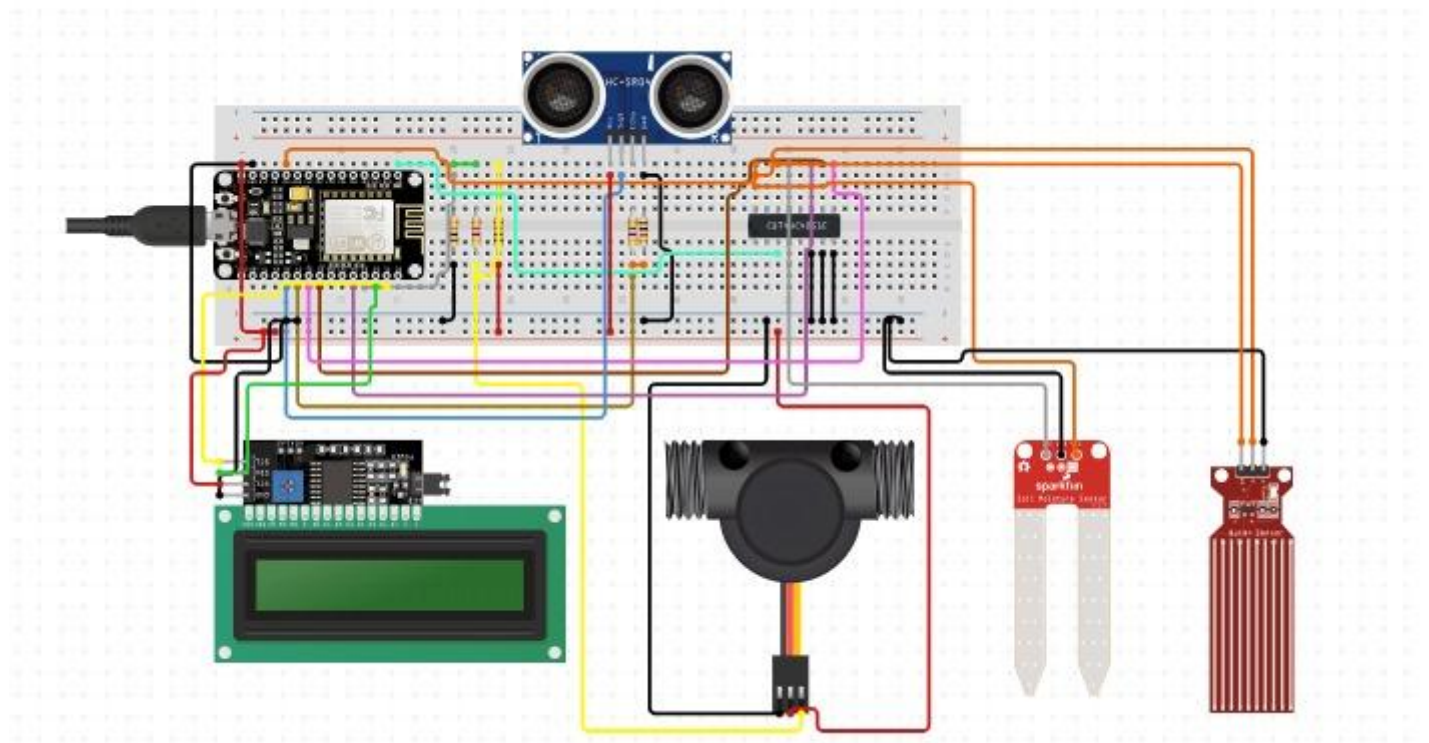


Figure 10.1 Circuit / Wiring Diagram

APPENDIX – 2

10.2 COMPONENTS SPECIFICATION

1. ESP8266 NodeMCU Microcontroller:

Controller Type: ESP8266 Wi-Fi microcontroller

Operating Voltage: 3.3 V

Digital I/O Pins: 17 GPIO pins

Application: Main controller for water-level, water-flow, and rainfall monitoring and wireless data transmission.

2. Ultrasonic Sensor – Water Level Detection:

Sensor Type: HC-SR04 ultrasonic sensor

Operating Voltage: 5 V

Output: Echo pulse signal

Application: Measuring the water level inside the drainage channel and identifying high-water or overflow conditions.

3. Water Flow Sensor:

Sensor Type: G1/2" Hall-effect flow sensor

Operating Voltage: 5 V

Output: Pulse signal

Application: Measuring water flow through the drainage channel and detecting reduced flow or no-flow conditions.

4. Rain Sensor:

Sensor Type: Rain detection sensor module

Operating Voltage: 3.3–5 V

Output: Analog/Digital

Application: Detecting rainfall conditions and supporting early flood-risk monitoring.

5. I2C LCD Display:

Display Type: 16 × 2 LCD

Interface: I2C

Operating Voltage: 5 V

Application: Displaying water level, flow rate, rainfall status, and warning information.

6. Water Level Sensor Module:

Sensor Type: Water level detection module

Operating Voltage: 3.3–5 V

Output: Analog signal

Application: Providing additional water-level information for monitoring and testing the drainage condition.

7. Power Supply:

Type: Regulated DC power supply

Output Voltage: 5 V

Function: Provides electrical power to the controller and sensors

Application: Supplying stable power to the ESP8266, sensors, and LCD display.

8. Drainage Prototype Channel:

Material: Lightweight fabricated material

Properties: Compact, rigid, and suitable for water-flow testing

Application: Represents a drainage channel for testing water flow, blockage, water-level rise, and overflow conditions.

9. Protective Sensor Enclosure:

Type: Waterproof/protective enclosure

Material: Plastic or lightweight enclosed material

Properties: Water-resistant and compact

Application: Protecting the ESP8266 and electronic connections from water and environmental exposure.

10. Software Monitoring Dashboard:

Type: Web-based IoT monitoring dashboard

Function: Real-time sensor data visualization

Features: Graphs, warnings, status monitoring, and CSV download

Application: Remote monitoring of water level, water flow, rainfall, blockage, no-flow, and overflow conditions.

APPENDIX – 3

10.3 PSEUDO CODE

The following pseudocode describes the main control logic of the Intelligent Drainage Blockage & Flood Prevention System. The program manages water-level measurement, water-flow monitoring, rainfall detection, blockage detection, overflow detection, LCD display, Wi-Fi communication, web dashboard monitoring, live graph updating, and CSV data logging.

The ESP8266 program is structured using the standard setup () and loop () functions. The setup () function initializes the I2C LCD, sensor pins, flow-meter interrupt, Wi-Fi connection, and web server. The loop () function continuously handles the web server, reads sensor values, calculates flow rate, updates the LCD, and stores monitoring history.

The program uses separate functions such as read Ultrasonic (), calculateWaterLevel (), read Sensors (), calculate Flowrate (), update LCD (), save History (), handle API (), and handle Download () to perform sensing, monitoring, display, dashboard communication, and data logging operations.

START

1. INITIALIZE SYSTEM

- Initialize ESP8266
- Initialize LCD
- Initialize Rain Sensor
- Initialize HC-SR04 Ultrasonic Sensor
- Initialize Water Flow Meter
- Initialize Digital Water Flow Detector

- Set sensor pins
- Set LCD address
- Set flow pulse interrupt

2. CONNECT TO Wi-Fi

- Start Wi-Fi connection

- IF Wi-Fi connected THEN

- Display IP address

- ELSE

- Display connection failure

- Restart ESP8266

END IF

3. START WEB SERVER

Initialize web server
Initialize dashboard
Initialize API data route
Initialize CSV download route

4. READ SENSOR DATA

Read rain sensor
Read digital water flow detector
Read ultrasonic sensor

IF rain sensor = LOW THEN
 Rain = YES
ELSE
 Rain = NO
END IF

IF water flow detector = LOW THEN
 Water Flow = YES
ELSE
 Water Flow = NO
END IF

5. CALCULATE WATER LEVEL

Measure distance using ultrasonic sensor

IF distance is valid THEN
 Calculate water level percentage
ELSE
 Water level = INVALID
END IF

6. CALCULATE FLOW RATE

Count flow meter pulses

Every 1 second:

- Calculate liters of water
- Calculate flow rate in L/min
- Calculate total water flow

7. DETERMINE DRAINAGE STATUS

IF Water Level $\geq$ 90% THEN

Status = OVERFLOW

Warning = DANGER

ELSE IF Rain = YES AND Water Level $\geq$ 70% THEN

Status = RAIN OVERFLOW

Warning = WARNING

ELSE IF Water Flow = NO AND Water Level $\geq$ 65% THEN

Status = BLOCKAGE

Warning = WARNING

ELSE IF Water Flow = NO THEN

Status = NO FLOW

Warning = WARNING

ELSE IF Water Flow = YES AND Water Level $<$ 65% THEN

Status = NORMAL FLOW

Warning = NORMAL

ELSE

Status = MONITORING

Warning = NORMAL

END IF

8. UPDATE LCD

Display water level

Display water flow status

Display flow rate

Display rain status

Display drainage status
Display distance
Display total water flow

9. SAVE MONITORING HISTORY

Every 5 seconds:

Save time
Save water level
Save distance
Save flow rate
Save rain status
Save water flow status
Save drainage status

Store maximum 30 records

10. UPDATE WEB DASHBOARD

Send sensor data to dashboard

Display:

Water Level
Distance
Flow Rate
Total Water
Water Flow
Rain Status
Drainage Status

11. DISPLAY LIVE GRAPH

Update water level graph
Update flow rate graph
Keep latest monitoring values

12. WARNING MONITORING

IF Overflow detected THEN
Display Overflow Warning

```
ELSE IF Blockage detected THEN
    Display Blockage Warning

ELSE IF No Flow detected THEN
    Display No Flow Warning

ELSE
    Display Normal Condition
END IF
```

13. CSV DATA DOWNLOAD

```
IF user selects CSV Download, THEN
    Generate monitoring CSV file
    Include sensor history
    Download CSV file
END IF
```

14. CONTINUOUS MONITORING

```
Repeat:
    Read sensors
    Calculate water level
    Calculate flow rate
    Determine status
    Update LCD
    Update dashboard
    Save history
    Check warnings

Continuously monitor drainage condition
```

END

CHAPTER 11

CONCLUSION

The Intelligent Drainage Blockage & Flood Prevention System was successfully designed and developed as a low-cost IoT-based drainage monitoring system. The main purpose of the project is to continuously monitor drainage conditions and provide early warnings when abnormal conditions such as water accumulation, blockage, no-flow conditions, or overflow occur. The system combines multiple sensors with an ESP8266 NodeMCU to collect and process real-time drainage data. The developed system uses an HC-SR04 ultrasonic sensor to measure the water level inside the drainage channel, while the water flow meter measures the flow rate. A digital water flow detector is used to identify whether water is flowing or not, and a rain sensor detects rainfall conditions. Based on the sensor readings, the ESP8266 processes the data and determines the drainage status. The system can identify conditions such as Normal Flow, No Flow, Blockage, Rain Overflow, and Overflow using predefined threshold values. For local monitoring, a 16×2 I2C LCD displays important parameters such as water level, flow rate, rainfall condition, water flow status, distance, and system status. In addition, the ESP8266 provides a Wi-Fi-based web dashboard for remote monitoring. The dashboard displays real-time sensor information along with live graphs for water level and flow rate. Warning information is also provided when critical conditions are detected, allowing the user to understand the drainage condition quickly. The prototype demonstrates that an IoT-based monitoring approach can reduce the dependence on continuous manual inspection of drainage systems. By monitoring water level, flow, and rainfall simultaneously, the system can provide useful information for identifying possible blockage and overflow conditions at an early stage. The collected monitoring data can also be downloaded in CSV format, which can be useful for further analysis and record keeping. Overall, the project provides a practical and economical solution for smart drainage monitoring and flood-risk prevention. The developed prototype successfully demonstrates the integration of sensors, microcontroller-based processing, LCD display, Wi-Fi communication, web dashboard, live monitoring, and warning functions. In future, the system can be further improved by adding cloud data storage, mobile notifications, additional drainage sensors, automatic alerts, solar power, and large-scale deployment across multiple drainage locations. Thus, the proposed system can serve as a foundation for developing a more advanced smart-city drainage and flood prevention network.

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